

NIKHILESH THIRUVENGADAM

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Education

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Science, Threads: Intelligence and Information Internetworks; GPA: 4.00/4.00

Graduation: May 2028

Experience

Scribur

Atlanta, GA

Co-Founder & Software Engineer

May 2026 – Present

- Architected a multi-tenant SaaS platform for in-house content teams on React, TypeScript, and Supabase (PostgreSQL).
- Enforced tenant isolation entirely at the database layer with **414 row-level security policies** across **142 tables**.
- Hardened tenant isolation with a 36-suite **adversarial harness (255 tests)** covering cross-tenant RLS matrices, prompt injection, MIME-spoofed uploads, payout abuse, and storage ACLs, plus Schemathesis fuzzing in CI.
- Shipped serverless Deno edge functions handling **Stripe billing**, **Twilio SMS**, and scheduled background jobs.
- Wrote an atomic sliding-window rate limiter as a single SQL RPC, closing concurrent-request bypass races.
- Engineered **ETL pipelines** pulling Instagram and TikTok engagement metrics through parallel Apify scrapers.

Agile Communication Architectures, Georgia Tech

Atlanta, GA

Undergraduate Researcher, AI-Driven Anomaly Detection for Network Systems

Aug 2025 – May 2026

- Implemented **zero-shot and continual learning** pipelines classifying unseen jamming signatures without retraining.
- Built preprocessing pipelines turning raw radio IQ samples into spectrograms and PCA embeddings for classifiers.
- Fused time-domain, frequency-domain, and autoencoder features to suppress false positives in live radio captures.
- Deployed trained models onto **USRP** software-defined radios for real-time jamming and spoofing detection in the field.

IBeeAnalytics

Cumming, GA

Software Engineering Intern

Aug 2023 – May 2025

- Developed and deployed **15+ production dashboards** in React across **20+ client sites** via versioned workflows.
- Refactored bespoke React templates into a **shared component library** adopted across all client site deployments.
- Automated **CI/CD deployments** with Vercel and AWS, enabling zero-downtime rollouts for client releases.

Projects

RefNet (2nd Overall Winner, HackGT 12) | React, Flask, Node.js (Express), D3.js, Docker, AWS EC2

- Rendered OpenAlex citation graphs with a D3 force layout measured at **553 ms median** for 1,000 nodes (~5K edges).
- Split the backend in two: a Flask API fetching papers from OpenAlex and an Express service running the AI assistant.
- Embedded a **GPT-4o assistant** that answers comparative questions across papers the user selects in the graph.
- Containerized Flask, Express, and React services with **Docker Compose** for single-command deployment to AWS EC2.

Terminal | Python, FastAPI, PostgreSQL, Redis, SQLAlchemy, WebSockets, Docker

- Designed an autonomous loop turning market news into trade theses, paper trades, and graded outcomes on PostgreSQL.
- Validated every financial figure the LLM cites against independently fetched market data before saving it.
- Graded every trade thesis on Brier score and calibration in a forward-only evaluation harness with **328 unit tests**.
- Unified 4 market-data APIs behind one interface with per-provider circuit breakers, documented in 5 design records.
- Gated market-wide news through an LLM triage pass dropping **~95%** as immaterial to stay inside free-tier API quotas.
- Ran two paper-trading accounts at 0.70 vs 0.50 confidence thresholds to test whether LLM confidence carries signal.

PolyTrader | Python, Flask, GitHub Actions, pytest

- Tracked **~150 top Polymarket wallets**, detecting when a trader actually sold a position versus when a settled market paid out automatically, and priced three-way markets where a single No bet covers two outcomes.
- Forward-tested each simulated bet against **market closing prices** with modeled slippage, backed by **51 unit tests**.
- Wrapped the Polymarket API in a **caching layer** with exponential backoff and a cap on parallel requests.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, Swift, Bash

Frameworks & Libraries: React, Node.js, Express, Flask, FastAPI, PyTorch, Pandas, NumPy, SQLAlchemy, D3.js

Tools & Platforms: PostgreSQL (Supabase), Redis, Docker, AWS (EC2), GitHub Actions, Git, Linux, Playwright

Concepts: Distributed Systems, Concurrency, Caching, RBAC, REST API Design, CI/CD, ETL, Machine Learning